

4.2.4 Energy transfers

4.2.4.1 Power

Content	Key opportunities for skills development
Students should be able to explain how the power transfer in any circuit device is related to the potential difference across it and the current through it, and to the energy changes over time: <i>power = potential difference × current</i> $[P = V I]$ <i>power = (current)² × resistance</i> $[P = I^2 R]$ power, P, in watts, W potential difference, V, in volts, V current, I, in amperes, A (amp is acceptable for ampere) resistance, R, in ohms, Ω	MS 3b, c WS 4.5 Students should be able to recall and apply both equations.

4.2.4.2 Energy transfers in everyday appliances

Content	Key opportunities for skills development
Everyday electrical appliances are designed to bring about energy transfers. The amount of energy an appliance transfers depends on how long the appliance is switched on for and the power of the appliance. Students should be able to describe how different domestic appliances transfer energy from batteries or ac mains to the kinetic energy of electric motors or the energy of heating devices. Work is done when charge flows in a circuit. The amount of energy transferred by electrical work can be calculated using the equation:	
$energy\ transferred = power \times time$ [$E = P t$] $energy\ transferred = charge\ flow \times potential\ difference$ [$E = Q V$] energy transferred, E, in joules, J power, P, in watts, W time, t, in seconds, s charge flow, Q, in coulombs, C potential difference, V, in volts, V	MS 3b, c Students should be able to recall and apply both equations. WS 1.4
Students should be able to explain how the power of a circuit device is related to: - the potential difference across it and the current through it - the energy transferred over a given time. Students should be able to describe, with examples, the relationship between the power ratings for domestic electrical appliances and the changes in stored energy when they are in use.	WS 1.2

4.2.4.3 The National Grid

Content	Key opportunities for skills development
The National Grid is a system of cables and transformers linking power stations to consumers.	

Content	Key opportunities for skills development
Electrical power is transferred from power stations to consumers using the National Grid. Step-up transformers are used to increase the potential difference from the power station to the transmission cables then step-down transformers are used to decrease, to a much lower value, the potential difference for domestic use. Students should be able to explain why the National Grid system is an efficient way to transfer energy.	The construction and operation of transformers is covered Transformers (HT only) (page 72). WS 1.4